

Python Lists, Tuples & Dictionaries Quiz

1. What keyword defines a function?

- A) def
- B) function
- C) lambda
- D) return

Correct Answer: A

Explanation: def is used to define functions (e.g., def foo():).

2. What is the output of:

```
def greet():  
    return "Hello"  
  
print(greet())
```

- A) None
- B) greet()
- C) "Hello"
- D) Error

Correct Answer: C

Explanation: The function returns the string "Hello".

3. Which is a valid function parameter?

- A) 1param
- B) param-1
- C) param_1
- D) global

Correct Answer: C

Explanation: Names can contain letters, numbers, and underscores (but can't start with numbers).

4. What does *args capture in a function?

- A) Keyword arguments
- B) Default values
- C) Positional arguments as a tuple
- D) Return values

Correct Answer: C

Python Lists, Tuples & Dictionaries Quiz

Explanation: *args collects extra positional arguments.

5. Which function call uses a keyword argument?

- A) func(1, 2)
- B) func(x=1, y=2)
- C) func(*[1, 2])
- D) func(1)

Correct Answer: B

Explanation: x=1 and y=2 are keyword arguments.

6. What is the default return value of a function without a return statement?

- A) 0
- B) ""
- C) None
- D) False

Correct Answer: C

Explanation: Functions return None by default.

7. Which is a valid lambda function?

- A) lambda x: return x*2
- B) lambda x: x*2
- C) def lambda x: x*2
- D) lambda (x): x*2

Correct Answer: B

Explanation: Lambdas implicitly return expressions (no return keyword).

8. What does map(lambda x: x2, [1, 2, 3]) return?**

- A) [1, 4, 9]
- B) (1, 4, 9)
- C) A map object
- D) Error

Correct Answer: C

Explanation: map() returns an iterator (convert to list with list()).

Python Lists, Tuples & Dictionaries Quiz

9. Which built-in function returns the number of items in an object?

- A) sum()
- B) count()
- C) len()
- D) size()

Correct Answer: C

Explanation: len() works for strings, lists, etc.

10. What does filter(lambda x: x>0, [-1, 0, 1]) return?

- A) [-1, 0, 1]
- B) [0, 1]
- C) A filter object containing [1]
- D) Error

Correct Answer: C

Explanation: filter() returns an iterator with elements where the lambda is True.

11. What is the correct way to import a module?

- A) import "math"
- B) include math
- C) import math
- D) from math import all

Correct Answer: C

Explanation: import module_name is the standard syntax.

12. How do you use sqrt from the math module without prefixing math.?

- A) import sqrt from math
- B) from math import sqrt
- C) import math.sqrt
- D) using math.sqrt

Correct Answer: B

Explanation: from module import name brings the name into the current namespace.

13. What does if __name__ == "__main__": do?

Python Lists, Tuples & Dictionaries Quiz

- A) Checks if the script is imported
- B) Runs code only when the script is executed directly
- C) Always runs the code
- D) Prevents the script from running

Correct Answer: B

Explanation: Used to separate executable code from importable code.

14. Which imports all names from a module?

- A) `import *` from module
- B) `import module.*`
- C) `from module import *`
- D) `import module.all`

Correct Answer: C

Explanation: This is generally discouraged (can pollute the namespace).

15. Where does Python look for modules by default?

- A) Only in the current directory
- B) In `sys.path` (includes current dir, installed packages, etc.)
- C) Only in `/usr/lib/python`
- D) Nowhere-you must specify full paths

Correct Answer: B

Explanation: `sys.path` lists all search locations.

16. What is `__init__.py` used for?

- A) To run a script automatically
- B) To mark a directory as a Python package
- C) To define global variables
- D) To create executables

Correct Answer: B

Explanation: Required in Python packages (even if empty).

17. What does this function do?

```
def foo(x, y=2):
```

Python Lists, Tuples & Dictionaries Quiz

return x * y

- A) Requires both x and y as arguments
- B) Allows y to be optional (default=2)
- C) Raises an error if y is not provided
- D) Returns x + y

Correct Answer: B

Explanation: y has a default value of 2.

18. What is the scope of a variable defined inside a function?

- A) Global
- B) Local to the function
- C) Available everywhere
- D) Depends on the variable name

Correct Answer: B

Explanation: Variables inside functions are local unless declared global.

19. Which is true about recursion?

- A) Python doesn't support it
- B) It requires a base case to avoid infinite loops
- C) It's always faster than iteration
- D) It doesn't use the call stack

Correct Answer: B

Explanation: Recursion needs a terminating condition.

20. What does nonlocal do?

- A) Accesses global variables
- B) Modifies variables in enclosing (non-global) scope
- C) Creates new variables
- D) Imports modules

Correct Answer: B

Explanation: Used in nested functions to modify outer function variables.